

|                                                                                   |                                                             |                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------|
|  | <b>DOCUMENT TITLE:</b><br>Suspension Quality Control Report | <b>Revision:</b><br>A               |
|                                                                                   | <b>DOCUMENT #:</b><br>TMP1-QC-C.SLQ001                      | <b>Effective Date:</b><br>23Sep2022 |

#### General Production Information

|                       |          |                                           |                                              |
|-----------------------|----------|-------------------------------------------|----------------------------------------------|
| <b>Work Order #:</b>  | 2302-01  | <b>NCMR #:</b>                            | N/A                                          |
| <b>Part Number:</b>   | C.SLQ001 | <b>Manufacture &amp; Expiration Date:</b> | Mfg date: 02/23/2023<br>Exp date: 02/23/2024 |
| <b>Part Revision:</b> | B        | <b>Batch / Lot #:</b>                     | 2302-01-2                                    |

#### Supplies Traceability

| Part Number | Description     | Part Revision | Lot / Batch # |
|-------------|-----------------|---------------|---------------|
| S.SLQ005    | pH Strip        | A             | 10BDH4801     |
| S.SLQ006    | Deuterium oxide | A             | M2338         |

#### Equipment Traceability

| Part Number | Description    | Equipment ID | Calibration Due Date |
|-------------|----------------|--------------|----------------------|
| E.SLQ013    | Weighing scale | ST-012       | 10/3/2023            |

#### QC Data

| Sample Tube | Weight Empty (g) | Weight with Residue (g) | Residue Net Weight (g) |
|-------------|------------------|-------------------------|------------------------|
| Sample A1   | 13.0751          | 13.1971                 | 0.1220                 |
| Sample A2   | 13.0260          | 13.1477                 | 0.1217                 |
| Sample B1   | 12.9875          | 13.1104                 | 0.1229                 |
| Sample B2   | 13.0291          | 13.1558                 | 0.1267                 |
| Sample C1   | 12.7829          | 12.9096                 | 0.1267                 |
| Sample C2   | 12.9327          | 13.0671                 | 0.1344                 |

|                                                                                   |                                                             |                                     |
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| Sample Tube                   | Weight Empty (g) | Weight with Sample suspended (g) | Weight with Residue on Evaporation (g) | Residue on Evaporation (%) |
|-------------------------------|------------------|----------------------------------|----------------------------------------|----------------------------|
| Suspension Test Sample        | 12.9996          | 44.3261                          |                                        |                            |
| Residue on Evaporation Sample | 13.0647          | 23.4921                          | 13.1620                                | 0.973%                     |

### Summary of Results

| Attribute              | Result                             | Acceptance criteria                                                                                                   | Meets Criteria? (circle one)               |
|------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Appearance (color)     | Opaque white                       | Opaque white                                                                                                          | <input checked="" type="radio"/> Yes    No |
| Appearance (form)      | Inviscid liquid                    | Inviscid liquid                                                                                                       | <input checked="" type="radio"/> Yes    No |
| Residue on Evaporation | 0.973%                             | 0.5% – 1.5%                                                                                                           | <input checked="" type="radio"/> Yes    No |
| pH                     | 6                                  | 5 – 8                                                                                                                 | <input checked="" type="radio"/> Yes    No |
| <sup>1</sup> H NMR     | Attach <sup>1</sup> H NMR spectra  | Identify Peaks with Amplitude >5% of Solvent Peak (4.75ppm)                                                           | <input checked="" type="radio"/> Yes    No |
| <sup>19</sup> F NMR    | Attach <sup>19</sup> F NMR spectra | Identify Peaks, if there is a peak at -76ppm determine approximate amplitude relative to peaks at -140ppm and -150ppm | <input checked="" type="radio"/> Yes    No |

Report completed by:  
Ethan Rao

Print name



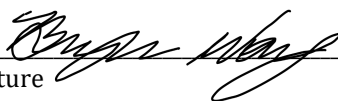
Signature

27 Mar 2023

Date

Report reviewed by:  
Bryan Wong

Print name



Signature

27 Mar 2023

Date



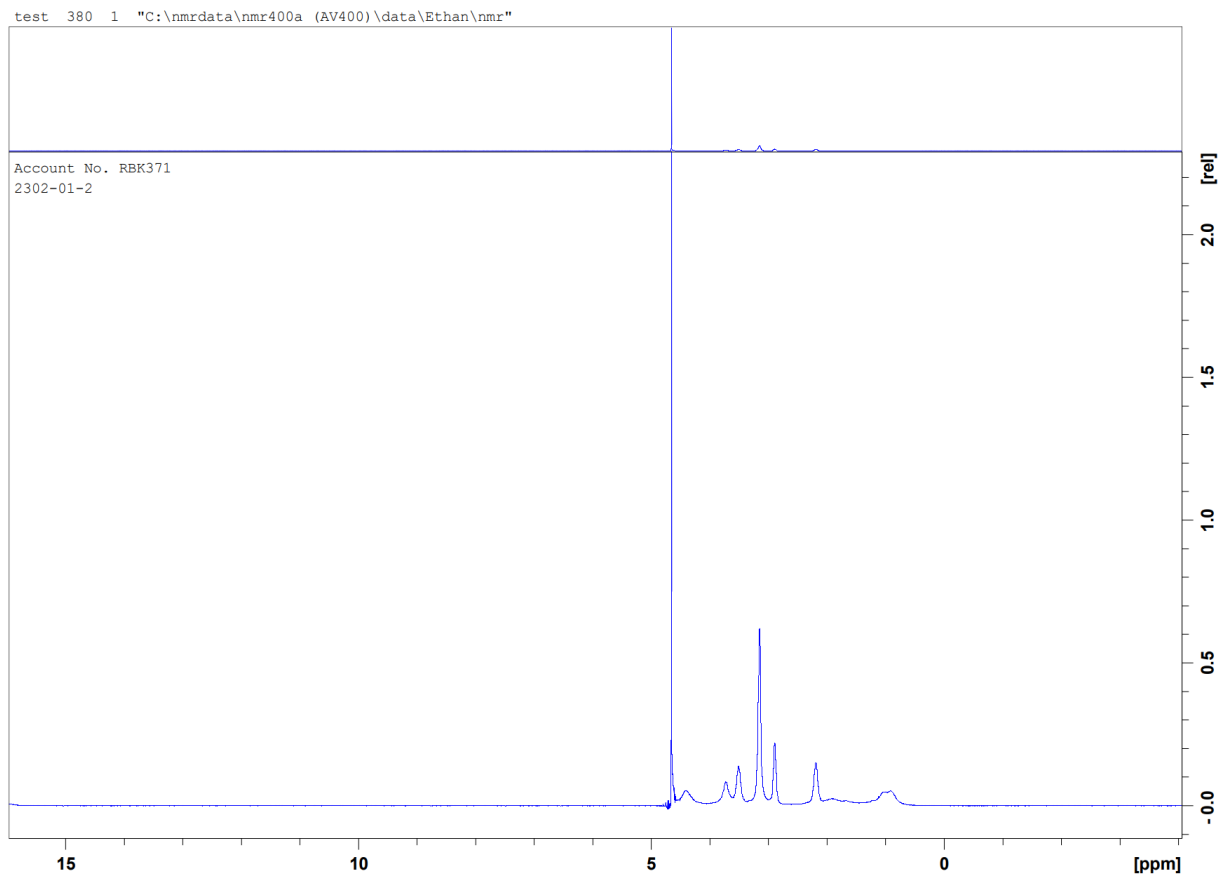
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**$^1\text{H}$  NMR:**



Broad Peaks are present at approximately 0.99, 2.2, 2.9, 3.1, 3.5, 3.7, 4.4, and 4.75 ppm indicating successful incorporation of the zwitterion moiety into the polymer.

**$^{19}\text{F}$  NMR:**



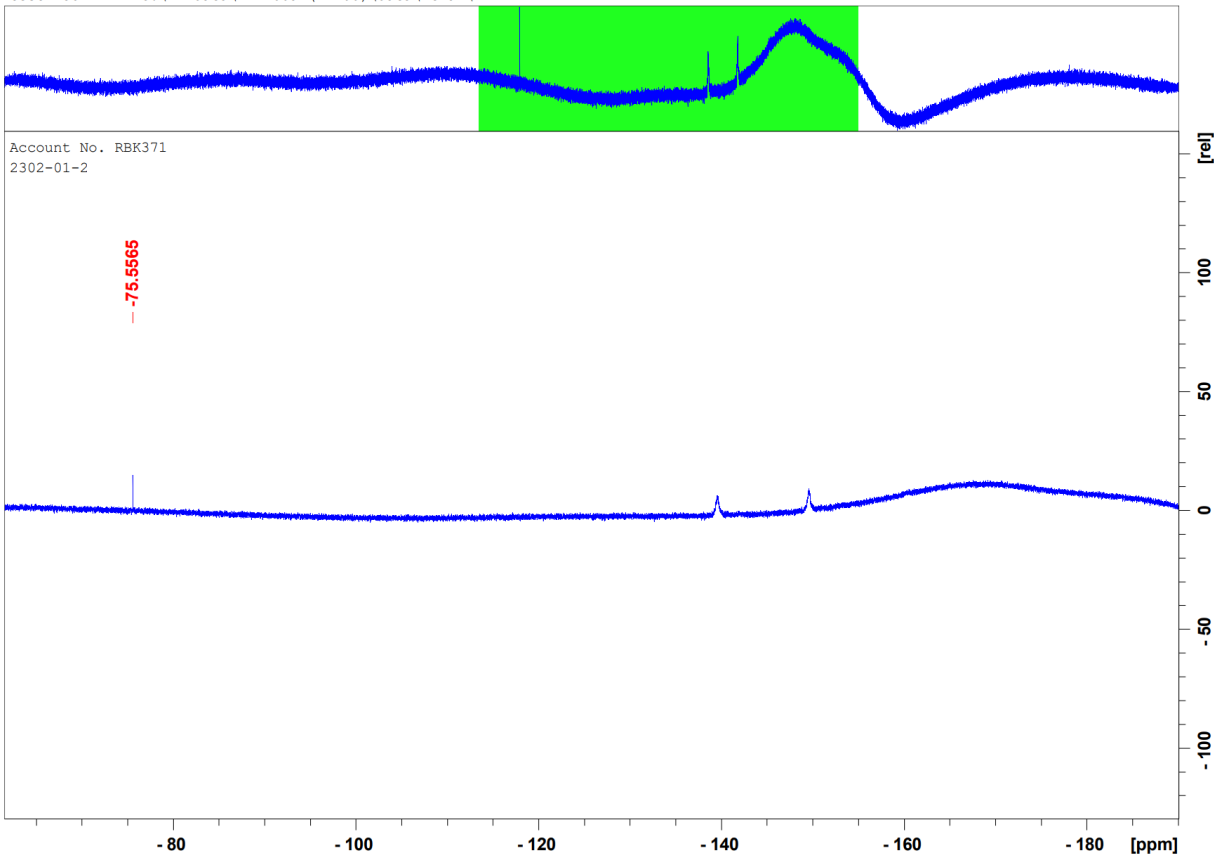
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test 381 1 "C:\nmrdata\nmr400a (AV400)\data\Ethan\nmr"



Two peaks at -140 and -150 ppm identified. One peak is identified at -75.5 ppm is due to the presence of trifluoroacetic acid which was added to the sample serving as a reference peak for the NMR spectra. No other peak was present at -76ppm indicating there is no TFE remaining in the suspension.